

# Electrodynamics of spin-1/2 particles

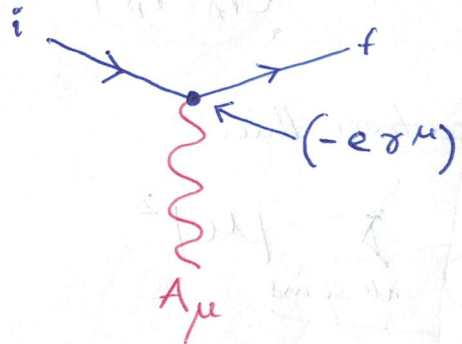
Electron current:  $(j_e^\mu)^{fi} = -e \bar{\psi}_f \gamma^\mu \psi_i$

$$\psi = u(p) e^{-ip \cdot x}$$

So, an electron would interact with an electromagnetic field  $A_\mu$  as

$$T_{fi} = -i \int (j_e^\mu)^{fi} A_\mu d^4x$$

$$= -i \int (-e \bar{\psi}_f \gamma^\mu \psi_i) A_\mu d^4x$$

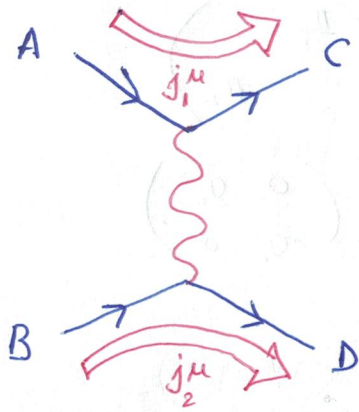


Remember in the spinless case

$$(j_e^\mu)^{fi} = -e (p_f + p_i)^\mu e^{i(p_f - p_i) \cdot x}$$

So we can now compute scattering of spin-1/2 particles.

# Møller Scattering $e^-e^- \rightarrow e^-e^-$



$$T_{fi} = -i \int j_{\mu}^{(1)}(x) \left(-\frac{1}{q^2}\right) j^{(2)\mu}(x) d^4x$$

$$= -i \int (-e \bar{\psi}_C \gamma_{\mu} \psi_A) \left(-\frac{1}{q^2}\right) (-e \bar{\psi}_D \gamma^{\mu} \psi_B) d^4x$$

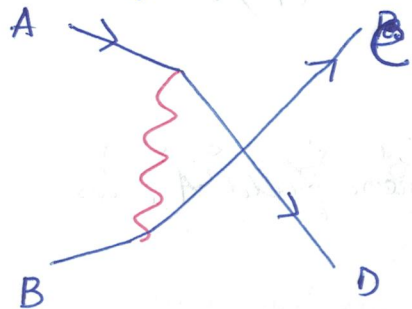
$\psi_i = u(p_i) e^{-ip_i \cdot x}$  → will give the delta functions

$$\therefore T_{fi} = -ie^2 \frac{(\bar{u}_C \gamma_{\mu} u_A)(\bar{u}_D \gamma^{\mu} u_B)}{q^2} (2\pi)^4 \delta^{(4)}(p_A + p_B - p_C - p_D)$$

$$= -i(2\pi)^4 \delta^{(4)}(p_{final} - p_{ini}) \mathcal{M}$$

$$\Rightarrow \mathcal{M} = (ie u_C \gamma_{\mu} u_A) \left(\frac{-ig^{\mu\nu}}{q^2}\right) (ie \bar{u}_D \gamma_{\nu} u_B)$$

Actually there is another diagram we need to consider



We can get the amplitude by interchanging

$C \leftrightarrow D$  + a '-' sign

(interchanging fermions)

So the total amplitude becomes

$$\mathcal{M} = -e^2 \frac{(\bar{u}_C \gamma^\mu u_A)(\bar{u}_D \gamma_\mu u_B)}{(P_A - P_C)^2} + e^2 \frac{(\bar{u}_B \gamma^\mu u_A)(\bar{u}_C \gamma_\mu u_D)}{(P_A - P_D)^2}$$

If we want to know the unpolarized cross-section then

$$|\mathcal{M}|^2 \rightarrow \overline{|\mathcal{M}|^2} = \frac{1}{\underbrace{(2s_A+1)(2s_B+1)}_{\text{averaging over initial spins}}} \sum_{\text{all spins}} |\mathcal{M}|^2$$

Let us analyze the non-rel. limit.  $|\vec{p}| \rightarrow 0$

$$\begin{array}{l} \text{Incoming } e^-: \quad U^{(s)} = \sqrt{2m} \begin{pmatrix} \chi^{(s)} \\ 0 \end{pmatrix} \\ \text{Outgoing } e^-: \quad \bar{U}^{(s)} = \sqrt{2m} \begin{pmatrix} \chi^{(s)\dagger} & 0 \end{pmatrix} \end{array} \quad \left| \quad \gamma^0 = \begin{pmatrix} \mathbb{I}_2 & \\ & -\mathbb{I}_2 \end{pmatrix} \right.$$

$$\gamma^{bi} = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}$$

$$\left[ \begin{array}{l} \bar{U}^{(s)} \gamma^\mu U^{(s)} = 2m \quad \text{if } \mu=0 \\ \quad \quad \quad = 0 \quad \quad \text{if } \mu=1,2,3 \\ \bar{U}^{(s)} \gamma^\mu U^{(s')} = 0 \quad \text{if } s \neq s'. \end{array} \right.$$

Spin cannot be flipped. (mostly electric interaction)

$$\mathcal{M}(\uparrow\uparrow \rightarrow \uparrow\uparrow) = \mathcal{M}(\downarrow\downarrow \rightarrow \downarrow\downarrow) = -e^2 4m^2 \left( \frac{1}{t} - \frac{1}{u} \right)$$

$$\mathcal{M}(\uparrow\downarrow \rightarrow \uparrow\downarrow) = \mathcal{M}(\downarrow\uparrow \rightarrow \downarrow\uparrow) = -e^2 4m^2 \frac{1}{t}$$

$$\mathcal{M}(\uparrow\downarrow \rightarrow \downarrow\uparrow) = \mathcal{M}(\downarrow\uparrow \rightarrow \uparrow\downarrow) = e^2 4m^2 \frac{1}{u}$$

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$$\therefore |\overline{M}|^2 = \frac{1}{4} (4m^2 e^2) \left[ \left( \frac{1}{t} - \frac{1}{u} \right)^2 + \frac{1}{t^2} + \frac{1}{u^2} \right]$$

In the center-of-mass frame

$$t = -2p^2(1 - \cos\theta) = -4p^2 \sin^2 \theta/2$$

$$u = 2p^2(1 + \cos\theta) = -4p^2 \cos^2 \theta/2$$

$$\frac{d\Omega}{d\Omega}_{cm} = \frac{m\alpha^2}{16p^4} \left( \frac{1}{\sin^4 \theta/2} + \frac{1}{\cos^2 \theta/2} - \frac{1}{\sin^2 \theta/2 \cos^2 \theta/2} \right)$$

